

PASSWORD SECURITY ANALYSIS AND POLICY DESIGN

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Tools Used: Python, Password Security Concepts

Platform: Windows / Kali Linux

Introduction

Passwords are one of the most common authentication mechanisms used to protect digital systems. Weak passwords can lead to unauthorized access, data breaches, and security incidents. Password security analysis helps identify weak passwords and improve authentication policies. In this project, password strength and security practices were analyzed to understand how strong password policies can enhance system security.

Objective

The objective of this project is to analyze password strength, identify weak passwords, and design a strong password policy to improve system security.

Tools Used

The tools used in this project include Python programming language and basic password security analysis techniques.

Methodology

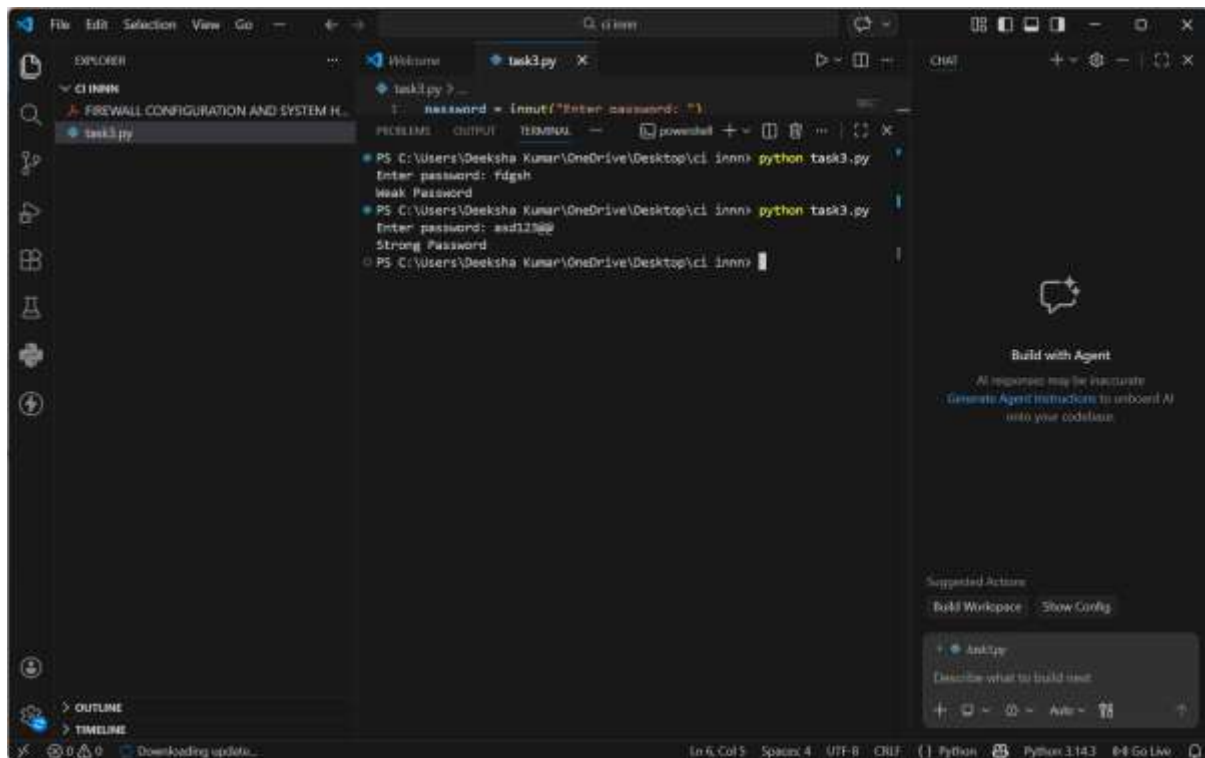
First, password security principles were studied to understand how attackers exploit weak passwords. Common weaknesses such as short passwords, predictable patterns, and reused passwords were identified.

Next, a simple password strength analysis was performed using Python to evaluate password length and complexity. Passwords were checked for the presence of uppercase letters, lowercase letters, numbers, and special characters.

Finally, a strong password policy was designed to ensure that users create secure passwords that are resistant to brute-force and dictionary attacks.

Results

The password analysis demonstrated that weak passwords such as short or predictable combinations can easily be exploited by attackers. Strong passwords that include multiple character types and longer lengths significantly improve security.



Security Analysis

Weak passwords are a major cause of security breaches. Implementing strong password policies, using multi-factor authentication, and encouraging password managers can significantly improve security.

Conclusion

Password security analysis helps organizations identify weak authentication practices and improve password policies. Strong password guidelines and proper authentication mechanisms are essential to protect systems from unauthorized access.